

Executive Project Report: End-to-End Loan Risk & Capital Analytics

1. Executive Summary

The primary objective of this capstone project was to analyze credit risk, underwriting patterns, and capital distribution for a retail banking portfolio consisting of 45,000 total loan applications. By building an interactive data pipeline across Excel, SQL, and Power BI, this dashboard transforms raw borrower profiles into strategic, executive-ready insights. The final product equips stakeholders with the visibility needed to optimize loan approval strategies, manage default exposures, and track overall portfolio yields at a glance.

2. Data Cleaning & ETL (Excel / SQL)

Before constructing the data model, extensive data preparation was conducted to ensure structural integrity and analytical accuracy:

- **Handling Nulls and Anomalies:** Identified and addressed missing values within critical fields such as `person_emp_length` (employment length) and `loan_int_rate` (interest rate) to prevent skewed calculations.
- **Data Standardization:** Formatted categorical variables (e.g., lowercase trimming, removing trailing spaces) for columns like `person_home_ownership` and `loan_intent` to ensure seamless aggregation.
- **Type Casting:** Ensured financial columns like `loan_amnt` (loan amount) were strictly designated as currency fields to maintain exact precision during high-volume data aggregation.

3. Data Modeling & DAX Engineering

Once imported into Power BI, the flat schema was structured into a clean data model, and custom Data Analysis Expressions (DAX) were engineered to drive the key metrics:

- **Total Applications** = `COUNT(loan_data[loan_id])` – Tracks total volume.
- **Average Interest Rate** = `AVERAGE(loan_data[loan_int_rate])` – Evaluates the portfolio's pricing strategy.
- **Total Funded Capital** = `SUM(loan_data[loan_amnt])` – Aggregates the absolute financial exposure across the organization.
- **Visual Enhancements:** Applied a professional design system featuring a light neutral canvas (`#F4F5F7`), a clear 4-tier horizontal KPI card ribbon, sharp white card visuals

with subtle drop shadows, and a unified deep navy (#1F4E79) and teal (#00A896) color palette.

4. Core Insights & Dashboard Analysis

The completed dashboard reveals three critical operational insights regarding the loan portfolio:

- **Risk Concentration in Housing Profiles:** A significant portion of the total funded capital is heavily concentrated among borrowers who list their housing status as **RENT** or **MORTGAGE**. Borrowers who rent present a distinctly different risk profile than traditional homeowners.
- **Capital Distribution by Intent:** Cross-analyzing **loan_amnt** by **loan_intent** indicates that categories such as **MEDICAL** and **EDUCATION** drive substantial application volumes, making them key anchors for the bank's consumer lending growth.
- **Portfolio Health at a Glance:** The centralized **loan_status** distribution allows credit risk managers to instantaneously monitor the ratio of performing loans against defaults, facilitating rapid shifts in credit underwriting policies when high-risk thresholds are breached.